



TECHNICAL GUIDE

MODELS: P*UR

**GAS-FIRED
CONDENSING / HIGH EFFICIENCY
UPFLOW FURNACES**

92 AFUE

**NATURAL GAS
37 - 130 MBH OUTPUT**



DESCRIPTION

These Category IV, highly efficient, compact, condensing type furnaces are designed for residential and commercial installations in a basement, closet, alcove, recreation room or garage where the ambient temperature is above 32°F, or higher. They may be either side wall or thru-roof vented using approved plastic type combustion air and vent piping. All units are factory assembled, wired and tested to assure dependable and economical installation and operation.

WARRANTY

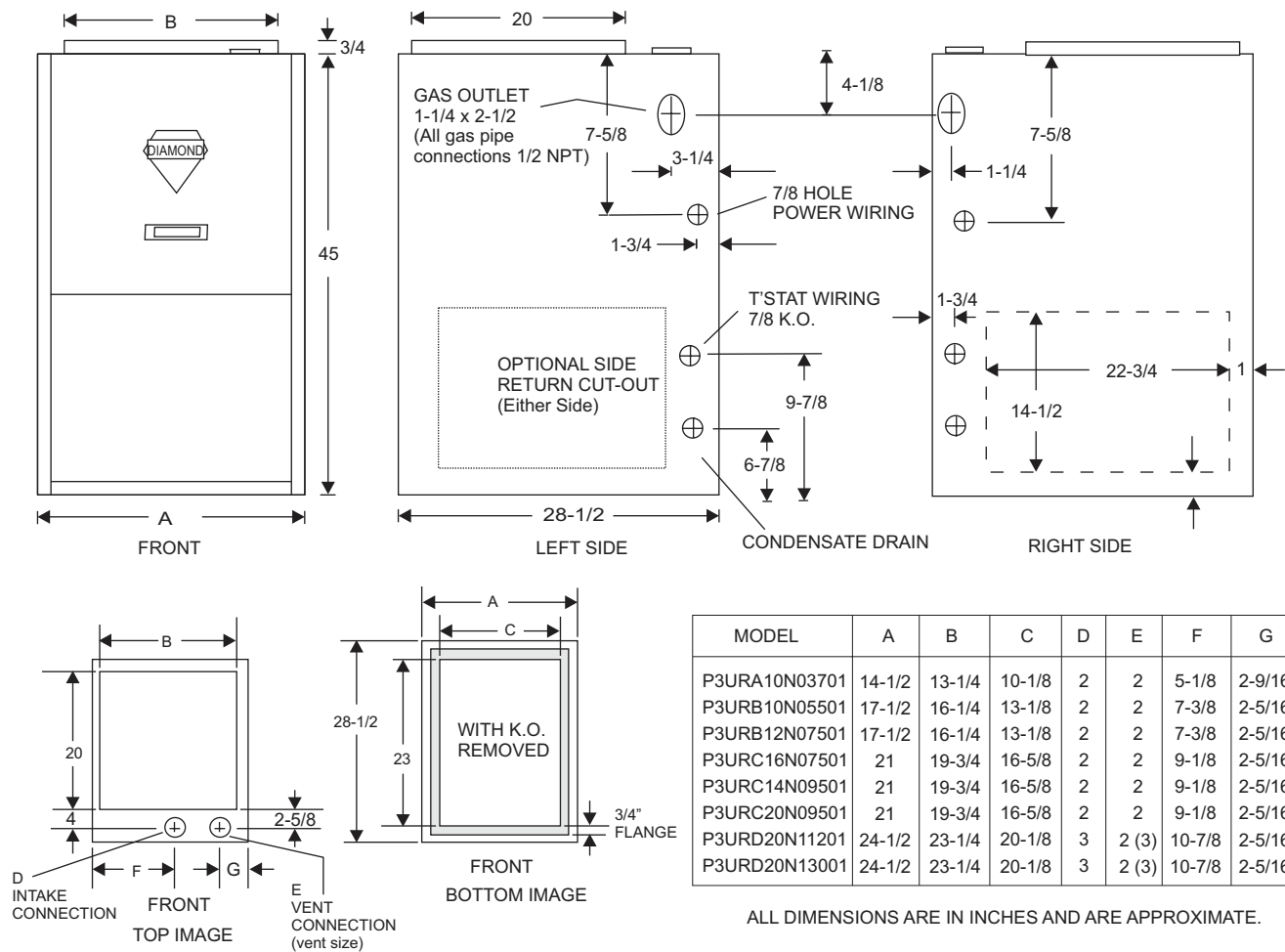
Lifetime limited warranty on both heat exchangers to the original purchaser; a 20-year limited warranty from original installation date to subsequent purchaser.

10-year warranty on the heat exchanger in commercial applications.

5-year limited parts warranty.

FEATURES

- Compact, easy to install, ideal height 45" cabinet
- Full height blower compartment allows side return application without the use of a return air box.
- Blower-off delay for cooling SEER improvement.
- Easy to connect power/control wiring.
- Built-in, high level self diagnostics with fault code display.
- Low unit amp requirement for easy replacement application.
- Integrated control module for reliable, economical operation.
- "Diamond Tough" hot surface ignitor.
- May be installed as either two-pipe (sealed combustion) or single pipe vent (using indoor combustion air)
- Top intake & vent connection allows installation in narrow locations.
- Electronic Hot Surface Ignition saves fuel cost with increased dependability and reliability.
- Induced combustion system with inshot main burners for quiet, efficient operation.
- No special vent termination kit required.
- 100% shut off main gas valve for extra safety.
- PSC - multiple speed, direct drive motor with large, quiet blower.
- 24V, 40 VA control transformer and blower relay supplied for add-on cooling.
- Hi-tech tubular aluminized steel primary heat exchanger.
- Secondary (condensing) heat exchanger of 29-4C high-grade stainless steel.
- Timed on, adjustable off blower capability for maximum comfort.
- Insulated blower compartment
- Solid bottom with removable panel allows easy application.
- Easy access from front of unit for cleaning, maintenance or service.
- Protection from intake, exhaust or condensate blockage.
- High velocity filter provided for field installation.
- Approved for mobile home application.



INTAKE/VENT SIZING TABLE - 1 PIPE / 2 PIPE SYSTEM

MODEL	PIPE SIZE	MAX. ELBOWS VS. ONE WAY VENT LENGTH (FT.)			
		5-40	45	50	75
P3URA10N03701	2"	6	5	4	N/A
P3URB10N05501					
P3URB12N07501					
P3URC16N07501					
P3URC14N09501					
P3URC20N09501					
P3URA10N03701	3"	8	7	6	5
P3URB10N05501					
P3URB12N07501					
P3URC16N07501					
P3URC14N09501					
P3URC20N09501					
P3URD20N11201	3" Only	6	5	4	N/A
P3URD20N13001					

HIGH ALTITUDE PRESSURE SWITCH APPLICATION

INPUT (MBH)	OUTPUT (MBH)	2,000 FT TO 6,000 FT.	> 6,000 TO 10,000 FT.
UPFLOW MODELS			
40	37	1PS0310	1PS0309
60	55	1PS0307	1PS0307
80/1200	74	1PS0306	1PS0306
80/1600	76	1PS0308	1PS0307
100	93	1PS0306	1PS0306
100	94	1PS0306	1PS0306
120	111	1PS0307	1PS0307
140	130	1PS0308	1PS0307

NOTE: For high altitude conversion, an orifice change may also be required. See Form 035-14460-001 for application information.

RATINGS AND PHYSICAL DATA

MODEL NUMBER	INPUT MBH	OUTPUT MBH	AFUE	AIR TEMP. RISE °F	CFM @ MEAN AIR TEMP. RISE	MAX. OUTLET AIR TEMP. °F	BLOWER			TOTAL UNIT AMPS	MAX.* OVER-CURRENT PROT.	MIN.* WIRE (AWG) @ 75'
							HP	AMPS	SIZE			
P3URA10N03701	40	37	92.4	35 - 65	685	165	1/4	6.20	9 x 6	9.0	20	14
P3URB10N05501	60	55	92.2	45 - 75	850	175	1/3	6.20	10 x 6	9.0	20	14
P3URB12N07501	80	74	92.0	35 - 65	1160	165	1/2	7.00	11 x 8	9.0	20	14
P3URC16N07501	80	76	94.3	30 - 60	1540	160	3/4	11.50	11 x 8	12.0	20	14
P3URC14N09501	100	93	92.2	45 - 75	1465	175	1/2	10.70	10 x 10	9.0	20	14
P3URC20N09501	100	94	93.0	30 - 60	1955	160	1	12.20	11 x 10	12.0	20	14
P3URD20N11201	120	111	92.0	45 - 75	1730	175	1	12.20	11 x 10	12.0	20	14
P3URD20N13001	140	130	92.0	45 - 75	2005	175	1	12.20	11 x 10	12.0	20	14

* Wire size and overcurrent protection must comply with the National Electric Code.

NOTES:

1. For altitudes above 2000 ft. reduce capacity 4% for each 1000 ft. above sea level.
2. Wire size based on copper conductors, 60°C, 3% voltage drop.
3. Continuous return air temperature must not be below 55°F.

MODEL NUMBER	FILTER SIZE		ADD-ON COOLING		APPROX. OPER. WEIGHT
	SIDE	BOTTOM	TONS	CFM* @ .5 ESP	
P3URA10N03701	16 x 25	14 x 25	1-1/2, 2, 2-1/2	1090	110
P3URB10N05501	16 x 25	16 x 25	1-1/2, 2, 2-1/2	1190	130
P3URB12N07501	16 x 25	16 x 25	2, 2-1/2, 3	1315	145
P3URC16N07501	16 x 25	20 x 25	3, 3-1/2, 4	1760	155
P3URC14N09501	16 x 25	20 x 25	2-1/2, 3, 3-1/2	1495	165
P3URC20N09501	(2) 16 x 25	20 x 25	3-1/2, 4, 5	2080	170
P3URD20N11201	(2) 16 x 25	22 x 25	3-1/2, 4, 5	2100	180
P3URD20N13001	(2) 16 x 25	22 x 25	3-1/2, 4, 5	2125	190

* ESP (External Static Pressure) .5" WG is at furnace outlet ahead of cooling coil.

NOTES:

1. All filters must be high velocity cleanable type.
2. Air flows above 1800 CFM require either return from two sides or one side plus bottom.

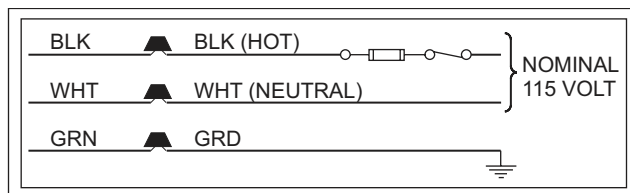
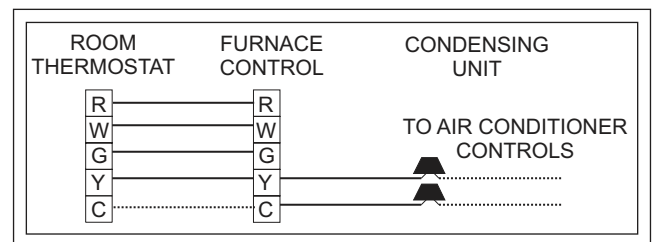
FIELD WIRING DIAGRAMS**POWER WIRING**

Table 1: BLOWER PERFORMANCE CFM- UPFLOW (WITHOUT FILTER)

MODELS	SPEED TAP	EXTERNAL STATIC PRESSURE, INCHES WC									
		0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
P3URA10N03701	HIGH	1330	1260	1210	1150	1090	1025	955	880	775	645
	MED-HIGH	985	970	965	935	900	860	800	740	665	535
	MED-LOW	790	780	760	740	715	685	635	575	485	365
	LOW	635	625	610	590	565	535	490	435	360	250
P3URB10N05501	HIGH	1360	1330	1290	1250	1190	1130	1055	970	865	735
	MED-HIGH	980	970	960	945	920	870	830	775	685	575
	MED-LOW	765	755	740	725	710	680	645	590	510	405
	LOW	635	625	610	590	565	540	500	440	375	290
P3URB12N07501	HIGH	1465	1440	1390	1365	1315	1255	1190	1125	1055	940
	MED	1085	1075	1060	1045	1030	1000	960	910	830	755
	LOW	805	795	785	770	755	730	700	655	615	565
P3URC16N07501	HIGH	1940	1905	1870	1815	1760	1700	1620	1555	1460	1305
	MED	1400	1390	1380	1370	1355	1320	1285	1255	1210	1135
	LOW	1245	1225	1205	1185	1165	1145	1125	1095	1050	970
P3URC14N09501	HIGH	1785	1720	1650	1575	1495	1405	1315	1190	1050	880
	MED	1430	1405	1375	1340	1290	1220	1130	1025	885	730
	LOW	1150	1135	1120	1080	1045	1000	925	825	700	575
P3URC20N09501	HIGH	2355	2285	2205	2145	2080	1995	1905	1820	1715	1610
	MED	1950	1915	1870	1830	1780	1710	1640	1560	1475	1375
	LOW	1575	1555	1540	1505	1470	1430	1380	1315	1245	1150
P3URD20N11201	HIGH	2375	2315	2245	2175	2100	2020	1935	1840	1755	1650
	MED	1990	1955	1920	1875	1835	1775	1715	1630	1550	1450
	LOW	1565	1550	1530	1505	1480	1445	1405	1350	1280	1200
P3URD20N13001	HIGH	2410	2340	2265	2200	2125	2035	1960	1860	1775	1670
	MED	2020	1975	1940	1895	1855	1790	1725	1655	1570	1470
	LOW	1580	1565	1550	1520	1495	1460	1420	1360	1295	1210
NOTE: Data below reflects airflows with two return openings - two sides or one side and bottom.											
P3URC20N09501	HIGH	2365	2295	2215	2155	2090	2005	1915	1830	1725	1620
	MED	1965	1930	1885	1845	1795	1725	1655	1575	1490	1390
	LOW	1595	1575	1560	1525	1490	1450	1400	1335	1265	1170
P3URD20N11201	HIGH	2385	2325	2255	2185	2110	2030	1945	1850	1765	1660
	MED	2005	1970	1935	1890	1850	1790	1730	1645	1565	1465
	LOW	1585	1570	1550	1525	1500	1465	1425	1370	1300	1220
P3URD20N13001	HIGH	2420	2350	2275	2210	2135	2045	1970	1870	1785	1680
	MED	2035	1990	1955	1910	1870	1805	1740	1670	1585	1485
	LOW	1600	1585	1570	1540	1515	1480	1440	1380	1315	1230
Airflow expressed in standard cubic feet per minute.											
Notes: 1. Return air is through side opposite motor (left side).											
2. Air flows above 1800 CFM require either return from two sides or one side plus bottom.											
3. Motor voltage at 115 V.											

FILTER PERFORMANCE

The airflow capacity data published in Table 1 on Page 4 represents blower performance WITHOUT filters. To determine the approximate blower performance of the system, apply the filter drop value for the filter being used or select an appropriate value from the Table 2 below.

NOTE: The filter pressure drop values in Table 2 are typical values for the type of filter listed and should only be used as a guideline. Actual pressure drop ratings for each filter type vary between filter manufacturer.

Table 2: FILTER PERFORMANCE - PRESSURE DROP INCHES W.C.

Airflow Range	Minimum Opening Size (in. ²)		Filter Type					
			Disposable		Hogs Hair*		Pleated	
	1 Opening	2 Openings	1 Opening	2 Openings	1 Opening	2 Openings	1 Opening	2 Openings
0 - 750	230		0.01		0.01		0.15	
751 - 1000	330		0.04		0.03		0.20	
1001 - 1250	330		0.08		0.07		0.20	
1251 - 1500	330		0.08		0.07		0.25	
1501 - 1750	380	658	0.14	0.08	0.13	0.06	0.30	0.17
1751 - 2000	380	658	0.17	0.09	0.15	0.07	0.30	0.17
2001 & Above	463	658	0.17	0.09	0.15	0.07	0.30	0.17

*. Hogs Hair Filters are the type supplied with furnace (if supplied).

APPLYING FILTER PRESSURE DROP TO DETERMINE SYSTEM AIRFLOW

To determine the approximate airflow of the unit with a filter in place, follow the steps below:

1. Select the filter type.
2. Select the number of return air openings or calculate the return opening size in square inches to determine the proper filter pressure drop.
3. Determine the External System Static Pressure (ESP) without the filter.
4. Select a filter pressure drop from the table based upon the number of return air openings or return air opening size and add to the ESP from Step 3 to determine the total system static.
5. If total system static matches a ESP value in the airflow table (i.e. 0.20, 0.60, etc.) the system airflow corresponds to the intersection of the ESP column and Model/Blower Speed row.
6. If the total system static falls between ESP values in the table (i.e. 0.58, 0.75, etc.), the static pressure may be rounded to the nearest value in the table determining the airflow using Step 5 or calculate the airflow by using the following example.

Example: For a 130,000 Btuh furnace with 2 return openings and operating on high speed blower, it is found that total system static is 0.58" w.c. To determine the system airflow, complete the following steps:

1. Obtain the airflow values at 0.50" & 0.60" ESP.
Airflow @ 0.50": 2125 CFM
Airflow @ 0.60": 2035 CFM
2. Subtract the airflow @ 0.50" from the airflow @ 0.60" to obtain airflow difference.
 $2035 - 2125 = -90 \text{ CFM}$
3. Subtract the total system static from 0.50" and divide this difference by the difference in ESP values in the table, 0.60" - 0.50", to obtain a percentage.
 $(0.58 - 0.50) / (0.60 - 0.50) = 0.8$
4. Multiply percentage by airflow difference to obtain airflow reduction.
 $(0.8) \times (-90) = -72$
5. Subtract airflow reduction value to airflow @ 0.50" to obtain actual airflow @ 0.58" ESP.
 $2125 - 72 = 2053$

ACCESSORIES

PROPANE (LP) CONVERSION KIT -

1NP0347 - All units except 130 MBH

1NP0349 - 130 MBH unit

This accessory conversion kit may be used to convert natural gas (N) units for propane (LP) operation. Conversions must be made by qualified distributor or dealer personnel.

EXTERNAL SIDE RACK W/FILTER - 1SR0302BK (6-Pack)

Includes rack that mounts to side of furnace cabinet and cleanable filter.

CONCENTRIC VENT TERMINATION -

1CT0302 (2")

1CT0303 (3")

For use through rooftop, sidewall. Allows combustion air to enter and exhaust to exit through single common hole.

CONDENSATE NEUTRALIZER KIT - 1NK0301

Neutralizer cartridge has a 1/2" plastic tube fittings for installation in the drain line. Calcium carbonate refill media is also available from the Source 1 Parts (p/n 026-30228-000).

EXTERNAL BOTTOM RACK W/FILTER -

Provides a cleanable, high velocity type filter and rack. Attaches to the bottom of the furnace.

1BR0314 - For 14" cabinets

1BR0317 - For 17-1/2" cabinets

1BR0321 - For 21" cabinets

1BR0324 - For 24-1/2" cabinets

HIGH ALTITUDE PRESSURE SWITCHES -

Used to convert units for operation at altitudes from 4,500 ft. to 10,000 ft. Refer to table on page 2 for proper pressure switch application. Refer to Form 035-14460-001 for high altitude application details.

NOTES

